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Ethics and Laws: Governing Generative AI's Role in Financial Systems

Prakriti Dixit Porwal

Department of Management Studies, Geetanjali Institute of Technical Studies, Udaipur, Rajasthan, India

"AI is probably the most important thing humanity has ever worked on"

– Sundar Pichai- CEO of Google and Alphabet

Abstract

Since generative artificial intelligence (AI) has evolved into more prevalent in financial systems, ethical and compliance concerns have emerged as vital parameters in its implementation. The present chapter discusses the complex relationship between generative AI tools with the financial arena, with a particular emphasis on the need for ethical principles and legal structures. The study highlights the potential effects of AI-generated outputs such as false dissemination, market forecasting, and algorithmic trading strategies, all of which have direct implications for financial stability, consumer protection, and data privacy.

Furthermore, the chapter underscores the complex challenges connected with inbuilt preconceptions, accountability opacity, transparency inadequacies, and unanticipated consequences that have emerged in real-world circumstances. To address these issues, a comprehensive framework is provided, one that smoothly integrates ethical foundations with regulatory processes to promote the secure and lawful integration of generative AI inside the banking domain.

The system demands continuous surveillance, periodic audits, and synergy among AI programmers, financial firms, and regulatory bodies. It seeks to capitalize on the benefits of generative AI while maintaining the integrity of

Email: prakritiporwal@gmail.com

Pethuru Raj Chelliah, Pushan Kumar Dutta, Abhishek Kumar, Ernesto D.R. Santibanez Gonzalez, Mohit Mittal and Sachin Gupta (eds.) Generative Artificial Intelligence in Finance: Large Language Models, Interfaces, and Industry Use Cases to Transform Accounting and Finance Processes, (283–298)

financial markets and protecting the interests of all stakeholders through a balanced approach. By following this balanced path, the chapter hopes to traverse the growing terrain of generative AI in finance, focusing on its transformational potential while adhering to ethical standards and regulatory compliance.

***Keywords*:** Generative AI, finance AI applications, ethical principles, legal and regulatory framework

Introduction

The advancement of technology in finance has been tremendous. It began with manual paper-based transactions and progressed to digital banking, internet trade, artificial intelligence (AI)-driven analytics, blockchain for safe transactions, and fintech innovations such as mobile payments [13]. In this volatile financial landscape, technological advancements have constantly transformed the way we do business, manage investments, and assure the sustainability of the economy. In the most recent developments, Generative AI is establishing itself as a potent force, drastically affecting the mode of operation of financial institutions. As generative AI becomes

Source: https://images.app.goo.gl/F7Mg876N3s8ywhTx9 more prevalent in the field of finance, it has brought attention to the urgent requirement for comprehension of the complex integration of ethics and laws controlling its implementation [3].

This chapter delivers a comprehensive look at the complexities that result from integrating generative AI into financial systems. It emphasizes the vital significance of ethical principles and legal frameworks in guiding the adoption of this transformative technology. In an era when AI-generated outputs have power over crucial components such as information distribution, market predictions, and algorithmic trading methods, it is critical to understand the direct and far-reaching ramifications for financial stability, consumer protection, and data security [10]. Moreover, it explores the complex concerns surrounding the use of AI in finance, such as biases, accountability, transparency issues, and unexpected outcomes [1]. To address these issues, it presents a system that combines ethics and regulations.

The aforementioned structure stipulates constant monitoring, frequent audits, and a constructive synergy between AI programmers, financial organizations, and regulatory bodies. The main goal is to capitalize on the evident benefits of generative AI while protecting the integrity of financial markets and the interests of all stakeholders. This chapter attempts to traverse the developing landscape of generative AI within the sphere of finance, striking a difficult balance between scientific advancement and ethical responsibility [4].

Applications of AI in Financial Systems

The pervasive influence of AI is visible across different industries, including finance and banking, in today's rapidly evolving economy. These industries are at the forefront of realizing AI's transformational power. Whether it is using chatbots to inquire about opening savings accounts or having banks perform proactive credit card activity verification, AI is changing the way customers interact with financial services. This evolution reflects the broader trend of AI transforming established practices, promising improved productivity, safety, and enhanced consumer experiences [20]. According to Insider Intelligence's AI in Banking research, 80% of banks recognize the significant benefits that AI provides. Furthermore, a large number of institutions are proactively preparing to integrate AI-powered solutions. According to UBS Evidence Lab research, 75% of respondents from banks with assets exceeding US$100 billion are currently implementing AI strategies. This figure compared to 46% among banks with less than

US$100 billion in assets, suggesting a significant discrepancy in the adoption of AI methods based on financial institution size.

AI in finance provides numerous advantages:

- Personalization: The process of tailoring services and products to specific users.
- Opportunity Generation: Determines new prospects through data analysis.
- Risk and Fraud Management: Enhances security through real-time monitoring.
- Transparency and Compliance: Ensures regulatory compliance and transparency.
- Cost Reduction: Automates tasks, improving operational efficiency and lowering costs.

AI has found several uses in the financial industry, transforming how financial institutions function, make selections, and provide customer service. Listed here are key AI applications in the realm of finance [5].

- Fraud Detection and Prevention: AI assists financial institutions in identifying and preventing fraudulent actions by analyzing transaction data and consumer behavior patterns. Machine learning algorithms can detect abnormalities and flag suspect transactions, lowering financial losses due to fraud.
- Customer Service and Chatbots: AI-powered chatbots and virtual assistants provide 24/7 customer service, answer inquiries, and help with mundane chores, improving the customer experience while lowering operating expenses. HDFC Bank, for example, employs “Eva,” a virtual assistant that answers customer questions, facilitates transactions, and offers information about account balances and previous transactions via their website and mobile app. Similarly, ICICI Bank uses “iPal” for customer service, answering questions, and assisting with account-related chores. These chatbots improve user experiences and offer clients in India rapid, round-the-clock support.
- Algorithmic Trading: AI-powered algorithms that analyze massive datasets and market circumstances in real time. They execute transactions faster and more efficiently than humans, capitalizing on market opportunities

while successfully limiting risks [16]. The National Stock Exchange's SMART Order Routing system, which optimizes trading execution and risk management, is one example.

- Credit Scoring and Underwriting: AI-powered Credit Scoring and Underwriting use a variety of data sources, including social media and online behavior, to improve credit ratings. This improves decision precision and broadens credit availability, especially for underprivileged groups, by providing fairer and more equitable lending possibilities to both individuals and enterprises. Companies like Upstart, for example, use AI to assess borrowers beyond FICO ratings, democratizing lending and allowing more people, particularly those with limited credit history, to access loans at reasonable conditions.
- Cybersecurity: AI contributes to strengthening financial system vulnerability. AI systems constantly track network traffic, employing machine learning algorithms to detect and mitigate cybersecurity risks in real time [16]. In this vein, AI-powered intrusion detection systems (IDSs) are capable of identifying irregular patterns and illicit access and promptly initiate responses by blocking hostile IP addresses. Furthermore, AI assists in fraud detection by analyzing large datasets to spot anomalous transaction behaviors, eliminating financial fraud, and safeguarding sensitive consumer data.
- Risk Management: AI has reshaped risk management, including credit, market, and operational risk assessments. JP Morgan, for example, employs advanced AI technology to quickly identify possible hazards in large datasets. This enables them to draw rational decisions as well as effectively deploy resources, thereby enhancing the financial sector's resilience and stability.
- Natural Language Processing (NLP): NLP is the process of extracting information from unstructured text data such as news articles and customer comments. NLP, for instance, Bloomberg's language-processing AI, is used by financial professionals to break down relevant information for sound investment choices. This technique aids in the navigation of large textual databases, hence improving financial decision-making processes [19].

- <u>Regulatory Compliance</u>: AI has altered regulatory compliance in the financial sector by streamlining activities such as transaction monitoring, anti-money laundering (AML) inspections, and know your customer (KYC) procedures. For example, HSBC uses AI-powered tools to analyze massive quantities of transactions in real time, detecting suspicious transactions and assuring compliance with tough AML laws. These automated solutions reduce compliance risks, improve accuracy, and speed up the ever-changing regulatory compliance process for financial institutions, resulting in a safer and more efficient industry landscape.

Ethical Challenges [5, 8, 12]

The incorporation of AI into financial systems has yielded never-before-seen efficiency and insights. However, technology also raises important ethical quandaries, such as worries regarding the following:

- <u>Transparency and Comprehensibility</u>: The difficulty is that a lack of openness in AI decision-making might impede understanding and responsibility. Many AI algorithms used in financial systems are "black-box" models, which means they make judgments without providing obvious reasons. Because of this lack of transparency, regulators, customers, and even the institutions that use these models may struggle to understand why a certain choice was reached [1]. High-frequency trading algorithms, for example, frequently operate as black boxes, making split-second trading decisions with no clear reason. Concerns have been expressed concerning market stability and fairness as a result of this lack of openness.
- <u>Financing Bias</u>: Prejudices in training data can be passed on to AI algorithms, resulting in biased lending behaviors. Credit scoring and underwriting algorithms powered by AI frequently rely on historical data, which may contain preconceptions based on ethnic background, sexual orientation, or socioeconomic considerations. When AI perpetuates these biases, it can lead to unfair lending practices. In 2019, it was shown that the credit limits on the Apple Card

were regularly lower for women compared to men with similar financial records. This emphasized the possibility of gender bias in AI-driven credit judgments [8].

- Privacy Concerns: The collection and use of sensitive financial data for AI analysis raises issues about privacy. AI systems in finance require access to massive volumes of personal and financial data, such as transaction records, credit scores, and investment portfolios. To avoid unauthorized access and breaches, it is critical to ensure the privacy and security of this data. For instance, the Equifax data breach in 2017 exposed the personal information of millions of people, including their financial information. This breach highlighted the importance of strong cybersecurity safeguards in the financial sector.
- Market Manipulation and Algorithmic Trading: High-frequency trading algorithms have the potential to influence markets, triggering unethical behavior. Algorithmic trading has emerged as a major force in financial markets, with algorithms executing trades at breakneck speed. While this has the potential to improve market efficiency, it also raises the risk of market manipulation and flash collapses [2]. In the "flash crash" on May 6, 2010, the Dow Jones Industrial Average dropped nearly 1000 points in minutes before recovering. This episode was blamed on algorithmic trading, prompting concerns about market stability.
- Customer Profiling: AI-powered customer profiling may intrude on privacy and personal liberty. Financial institutions and technology companies frequently employ AI to generate comprehensive descriptions of customers for targeted marketing and product suggestions. While this can improve customer experiences, it also raises worries about intrusive data collecting and manipulation [9]. Retailers, for example, employ predictive analytics to identify and target weak consumers with aggressive sales practices, thereby leveraging their personal and financial vulnerabilities.
- Regulatory Compliance: The challenge is that the rapid progress of AI may outstrip regulatory frameworks, resulting in compliance gaps. The rapid growth of AI technology frequently outpaces authorities' ability to draught and enforce acceptable regulations and standards. As financial firms navigate a shifting regulatory landscape, this can present legal

and ethical issues. For example, the advent of cryptocurrencies and decentralized finance (DeFi) has generated regulatory issues for governments and financial institutions all over the world, as they try to figure out how to oversee these new financial innovations [6].

- Job Displacement: Automation of financial tasks may result in job displacement and economic inequity. The use of AI and automation in financial services may result in the replacement of human workers, particularly in routine and administrative positions. This raises ethical questions about employment loss and economic injustice. For example, the rise of robo-advisors, who utilize AI to handle investment portfolios, has reduced the need for traditional financial advisors in several areas.
- Sustainable Challenge: The high energy consumption of AI data centers poses environmental issues. AI models, particularly those employed for deep learning, demand a substantial amount of computer resources and energy. The carbon footprint of AI data centers, as well as the environmental impact of these operations, have generated worries about their long-term viability. For instance, many tech companies' data centers, which house AI infrastructure, have been chastised for their high energy use and carbon footprint.
- Financial Inclusion: While AI can improve financial services, it might also eliminate consumers who do not have access to digital technologies. Individuals and communities without Internet or digital devices may be excluded from AI-driven financial advances due to the digital gap. The asymmetrical availability of financial services poses ethical considerations. For example, rural or underprivileged communities with low internet access may be excluded from the benefits of AI-driven financial solutions, aggravating financial inequality.
- Ethical Investing: The difficulty is that AI-driven investing judgments may collide with ethical principles. AI-powered investment strategies may prioritize profitability above moral principles, eventually resulting in investments in industries or firms that disagree with the ethical ideals of an individual or organization [18]. For example, some AI-driven investment funds may mistakenly invest in businesses such as

fossil fuels or armament manufacturers, which contradicts investors' ethical or environmental ideals.

These ethical issues highlight the significance of confronting the responsible and ethical usage of AI in financial systems. To properly address these difficulties, financial institutions, regulators, technology developers, and society at large must work together to define clear standards, legislation, and ethical frameworks for AI applications in the financial sector. Here are some of the **Ethical and Legal Concerns associated with the use of AI in Finance** [11]:

- Transparent AI Decision-Making: To improve understanding and accountability, AI decision-making procedures must be transparent and comprehensible [1]. In one instance, a financial institution may use AI models to provide extensive explanations for lending decisions, making it clear to consumers and regulators why a loan was accepted or denied. This commitment to transparency fosters confidence and alleviates worries regarding hidden biases. Transparency should be a primary priority in AI-driven financial systems to foster stakeholder trust. When AI algorithms make decisions without explicit explanations, it becomes difficult for regulators, customers, and organizations that use these models to understand and manage these systems. This lack of openness can raise issues about justice, accountability, and ethical implications [15].
- Fair Financing and Bias Mitigation: It is critical to have methods established to detect and mitigate biases in AI algorithms used for lending decisions, providing fair and equitable access to financial services. As an illustration, a lending platform assesses its AI models for biases regularly and applies retraining strategies to remove biases in credit scoring algorithms. This proactive strategy aids in the prevention of unfair lending practices that may arise as a result of distorted training data. When assessing creditworthiness, AI systems frequently rely on past data, which can introduce biases due to factors such as ethnicity, age, gender, or social and economic standing. To address this ethical challenge, responsible financial institutions work aggressively to discover and correct flaws in their AI models, ensuring that loan choices are fair and non-discriminatory.

- Data Privacy and Security: Prioritizing the privacy and security of sensitive financial data used by AI systems is critical to prevent unauthorized access and potential breaches. To protect consumer financial data kept within AI-driven systems, a bank, for example, uses advanced encryption techniques, strict access limits, and regular security audits. This strong approach ensures that consumer information is kept private and secure. To power AI systems, financial institutions rely largely on massive volumes of personal and financial data. Protecting this data is not just an ethical necessity, but is also a legal requirement in many areas. High-profile data breaches, such as the Equifax hack in 2017, highlight the critical significance of good cybersecurity safeguards in the financial sector.
- Regulation and Market Integrity: Financial institutions should work with regulatory bodies to develop and implement ethical AI practices, particularly in high-frequency and algorithmic trading. Financial authorities, for example, collaborate closely with trading businesses to develop norms and standards for algorithmic trading practices that prioritize both market stability and fairness, assuring ethical behavior in the marketplace. Although algorithmic trading has transformed financial markets, it also introduces hazards such as market manipulation and volatility. Responsible financial institutions and regulators work together to develop and implement ethical rules to protect market integrity and reduce unethical behavior.
- Ethical Customer Profiling: Ethical customer profiling entails the use of AI-powered approaches to improve interactions with customers while protecting their privacy and personal liberty. While customer profiling has the potential to improve service and overall experiences, it must be done ethically. Prioritizing privacy and giving clients control over their data is critical for appropriate AI-driven customer profiling [14]. A retail bank, for example, uses predictive analytics to construct customer profiles to make personalized product suggestions. They also adhere to strict data privacy regulations and give clients the choice to opt out of data collecting if they so desire.

- Robust Regulatory Compliance: Maintaining strict regulatory compliance in AI-powered financial systems is critical. It entails remaining proactive in addressing changing rules and ensuring AI systems adhere to legal norms. For example, a digital lending fintech keeps a specialized compliance staff on hand to constantly monitor and change its AI processes to suit evolving regulatory standards. The aforementioned commitment protects responsible lending practices and bridges the gap between rapid AI technology and dynamic regulatory landscapes, assuring ethical and legal adherence in financial operations [6].
- Job Displacement Mitigation: Mitigating job displacement caused by AI adoption requires responsible organizations to engage in worker reskilling and development programs [17]. A financial institution, for instance, provides thorough training to staff impacted by automation, allowing them to shift to different tasks within the organization. These programs address ethical concerns about unemployment and economic inequality, ensuring that employees remain adaptive and prepared for changing labor environments [7].
- Green AI Practices: To reduce the environmental impact of AI operations, responsible organizations prioritize energy-efficient AI technology and data centers. For example, a technology business may actively use energy-efficient gear and data centers powered by renewable energy sources for its AI operations, reducing the carbon footprint often associated with AI technology. These long-term projects are crucial since AI models, particularly those used in deep learning, can be computationally heavy and require significant energy resources, emphasizing the commitment to environmental sustainability.
- Financial Inclusion Projects: These entail responsible financial organizations producing AI-powered financial services that emphasize openness for all to bridge the digital gap. A digital bank, for case in point, may collaborate with nonprofit organizations to provide AI-powered mobile banking services in areas with poor internet connection, ensuring that underprivileged consumers have access to vital financial services. These activities highlight the dedication to ensuring that AI-driven financial solutions remain inclusive,

regardless of people's or communities' technology resources, encouraging financial inclusion.

- Ethical Investment Portfolios: Responsible financial institutions create AI-driven investment tactics that relate to ethical and environmental principles and allow investors the ability to invest in investments that adhere to their core values [18]. For example, an asset management business may offer AI-powered investment portfolios that exclude companies involved in areas that its clients consider unethical, which might include fossil fuels or arms manufacture. These initiatives represent the growing trend of ethical investing and demonstrate how AI may assist investments that align with investors' ethical and environmental objectives, fostering responsible financial practices [7].

The aforementioned instances demonstrate how the appropriate and ethical use of AI inside financial systems can effectively solve potential ethical concerns while providing consumers with important financial services. Adopting such practices demonstrates financial organizations' commitment to ethical standards and adds greatly to the development of confidence among their stakeholders [17].

Ethical AI in Indian Finance: Case Studies and Insights

- ICICI Bank, an acclaimed Indian financial institution, deployed AI-powered chatbots, including IBM's Watson, to enhance customer care interactions. These chatbots were created to help customers with their inquiries and transactions. However, their implementation prompted privacy and security problems. While ICICI Bank tried to improve client experiences through AI adoption, it was also scrutinized for its handling of sensitive customer data. This instance emphasizes the vital importance of integrating AI use in financial services with severe data protection legislation and ethical considerations to safeguard the security of client information and the maintenance of trust in AI-powered services [8].
- Paytm, a renowned Indian fintech company, has entered the realm of algorithmic trading, aided by AI-powered tools such as proprietary algorithms. While innovative, this venture

sparked regulatory concerns about market integrity and equity. The rapid spread of algorithmic trading heightened concerns about potential market manipulation [13]. The Paytm case emphasizes the importance of having broad and adaptable legislation to supervise AI's participation in trade. Such rules should address not only market manipulation but also transparency, accountability, and ethical concerns to ensure that AI-driven trading practices are consistent with fair market principles and foster investor confidence.

- HDFC Bank used AI-powered credit scoring methods, including machine learning algorithms, to evaluate loan applicants. However, situations have occurred in which these AI-driven judgments looked to be biased, particularly in lending choices. Such studies generated ethical questions about the objectivity of AI-driven credit ratings. This story starkly demonstrates the vital importance of proactively tackling bias in AI algorithms to guarantee that lending practices are equitable and free of discrimination. It emphasizes the responsibilities of financial institutions to regularly evaluate and fine-tune their AI models to promote fairness and openness in credit evaluation processes, hence sustaining ethical norms in financial services.
- Aditya Birla Capital uses AI, including NLP algorithms, to give clients personalized investment advice and portfolio management services. While this AI integration promised to improve consumer experiences by providing personalized financial advice, it also highlighted ethical concerns about the privacy and security of sensitive financial data. The case emphasizes the critical need for comprehensive and up-to-date data protection legislation in the financial sector. These legislations should not only protect client information but also create clear guidelines for the ethical use of AI in financial portfolio management, balancing innovation with data protection.
- Zerodha, a renowned Indian brokerage firm, has integrated AI, including machine learning algorithms, into its trading platform, providing traders with AI-powered tools to help them make investing decisions. While the goal of this innovation was to empower traders, it also sparked important discussions about accountability, transparency, and market integrity. The use of AI in trading operations generated

concerns about the fairness of automated trading and the possibility of market manipulation [2]. It highlights the need for regulatory monitoring and the need for ethical trading practices to guarantee that AI-powered trading platforms run transparently and respect market integrity, hence sustaining investor and regulator trust.

Conclusion

AI is transforming the realm of finance through enhanced accessibility as well as efficacy for individuals and businesses alike. Chatbots and robo-advisors, for example, provide seamless customer experiences, while fraud detection and data analysis enable businesses to make informed decisions. This transformation is changing the way finance is accessed and managed. Finally, the incorporation of generative AI into the financial sector constitutes a significant step forward in technical growth, providing new prospects for efficiency and creativity. This transition, however, brings with it a slew of ethical and legal issues that necessitate careful analysis and proactive responses. The ethical challenges surrounding AI in finance are broad and multifaceted, ranging from transparency and bias mitigation to data privacy and market integrity. To appropriately traverse this changing landscape, financial institutions, regulators, technology developers, and society as a whole must work together to define clear norms, robust legislation, and ethical frameworks that regulate AI use in finance. Transparency in AI decision-making, equitable finance, and strong data privacy and security safeguards are critical foundations of responsible AI integration. Furthermore, tackling concerns such as employment displacement, environmental sustainability, financial inclusivity, and ethical investment portfolios helps to guarantee that AI-driven financial services stay inclusive and in line with social values. The instances cited from Indian financial institutions highlight the importance of a balanced strategy that incorporates technology innovation as well as ethical issues. By proactively and transparently addressing these concerns, the financial sector may reap the benefits of AI while protecting customer interests, market integrity, and responsible finance principles.

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